Laboratory Procedures

One primary goal of the course (other than specific content) is to get you to think, perform experiments, and communicate in a professional manner as a researcher or primary investigator. The style of writing and experimentation is as a professional research experience!

I will assign particular tasks for each lab assignment (task sheets). I will demonstrate basic procedures and use of equipment for each laboratory experiment. However, since the class may be doing several experiments simultaneously you will be expected to put in your own independent efforts. You will work with a lab partner that will rotate (you may not have the same lab partner twice throughout the course). I will be assigning lab partners. You are expected to learn the equipment through use and to be able to explain operations on your own. Since in some cases we may have different models or brands being used ---I will NOT explain all "push button" operations, but will explain basic functions –e.g. oscilloscopes measure Voltage vs. Time, and have operations such as waveform averaging of multiple traces. You are expected to figure out how to do what is needed! You may ask me or consult with each other. Looking up manuals is useful (I link those I have available).

Your data taking will require that you spend time on your own in lab (with your lab partner). You must work with a lab partner on these experiments. Equipment availability may allow several students to work at the same time on some labs, for other labs time may need to be scheduled per sign up times due to equipment availability. Each group must have their own data, and lab book recordings. You are taking data with your lab partners and performing the experiment together. The write ups are your own (completely). I do expect that you each have a record of data that is used (or not used). Your data should be identical. A partner handing in different data than yours will result in both lab grades being penalized (there can be only one raw data set and only one resulting final value –excepting significant figures and rounding issues). Get together with your partner on the analysis and final numbers---- **DO NOT COLLABORATE ON THE WRITE UP. You are not permitted to share any electronic correspondence (no figures, no pictures, not pictures of handwritten work on the board, nothing).**

There are several components to each lab:

1) Analysis of data (50 pts). This is a guarantee that you are interpreting the experiment correctly and have taken data in a timely manner. You may consider this a rough draft of your lab report, but need not contain all components. For example in the speed of light lab you must have your experimental uncertainty, your line fit, your line fit uncertainty and sample data plot, data table. All the elements that lead to a result must be completed and must be correct. Format on the portions that are included here is graded (graphs will have properly labeled axes and titles and captions, likewise tables). Uncertainty is included and listed correctly. This will turn into the "Results/Analysis" section(s) of your final

report. This should be typed, where possible adhere to AJP format, include numbered equations etc (this will become a part of your final report)!!!!!!!!

2) Demonstration of Lab skills (25 pts), hands on one on one evaluation) Taking the data with a working set-up and be able to demonstrate—each person is responsible for demonstrating to me that they understand the setup. For example with speed of light I will adjust detectors, change oscilloscope settings, etc and ask that you quickly take two data points and that you calculate a rough speed of light. Other experiments will have specific demonstration tasks such as for the Michelson interferometer you will be asked to demonstrate white light fringes (after I completely mess up alignment).

3) Final Lab report (50 pts) You will have comments from me returned regarding format and any analytical issues from the Results. You will hand in the "rough draft analysis" along with your final report. Lab report details are below, and sample AJP formatted papers are online. Expectations are that physics is correct, discussion is included, analysis, correct and appropriate communications, format of all items.

   Note that research is an open ended task. There is no such thing as "complete" or "done". Open ended questions will always remain and there will be someone who thinks of creative additional tasks or interpretations.

Following the final lab report there will be peer reviews. I am doing this differently than in the past. Rather than two reviews by each person and for each person's paper, I will ask for selected reviews to be done following final reports. In research settings, publications, grants applications, it would be typical to have colleagues review work prior to submission. Then it would be typical for agencies to review your submitted work and give feedback to you. It is too cumbersome for our review process to follow that format, but you will each complete one or two reviews. You are expected to be highly critical. Reviews will count as a portion of your homework grade.

Logistics and Timing:
In general you will have about one to two weeks to take lab data and to write up an analysis. This means you will likely be starting on a next lab while completing final draft for the last one. The final lab report will typically be due about a week after I get the analysis back to you. The hands on lab demonstration must be done ASAP in all cases. You should schedule this with me as soon as you have completed taking your data. Equipment may need to be dismantled in order to set up the next lab.

Lectures and lab time: I expect you to be there for all class and lab times. There are a variety of tasks that may be done in either setting. Lecture on laser safety and lab issues, lecture on uncertainty analysis, lecture on spectroscopy or other experimental background, work trouble shooting (your analysis and lab reports), lab demonstrations, etc. We may simply meet to hand in work and schedule hands on demo's. If I am not lecturing then you should spend time on this course writing, analyzing, etc. If we are not using the time for me to present formal materials---then you are free to have informal Q&A discussions with me or to work on reports, peer review, in lab or other course work---it is expected this time is spent on this course. Lab work cannot be done late since I will need to put away equipment and get next set out.

After we get rolling through course work, much of the regularly scheduled class and lecture time will be spent on logistics and peer review. These are critical. Do not miss lab demonstrations (this is where I show you how to do the lab). These are critical.

LABORATORY BOOKS: We will use the following lab notebook (https://www.amazon.com/gp/product/B00007LV4B/ref=oh_aui_detailpage_o00_s00?ie=UTF8&psc=1). Purchase at least one, though it is likely you will need a second before the end of the course. I will not micro manage your lab notebooks, but state there are many philosophies on this. Some believe that this must be maintained as a legal document, others that it is a guide to enable others to follow your work, or last that it is a document to assist you and you alone in keeping a record for yourself. My own philosophy falls into the latter category. I will check that you are able to use your lab notebooks. It will be the only item you can bring with you to your hands on lab check. You must leave room for a table of contents. You must have data and lab description in place (neatly) prior to your lab demonstration exercise.

Here is a good link on lab note books and lab report writing.
http://www.drjbloom.com/Public%20files/Laboratory%20Notebook%20Booklet.pdf
You should look at the lab grading paperwork to get a good idea for how I will be grading.

Each lab should have a separate section in your lab notebooks. . You must record data, date, conditions, any information used in your report. Additional reference material are permitted and recommended to be placed in the lab (taped or stapled) notebooks such as tables, figures/pictures of equipment, graphs etc. The work in lab notebooks is for your own use. Each page should be numbered, data should have times and dates, and notes on procedures and equipment. This means that if you want to go back in five years and duplicate an experiment, the information necessary for you to do this should be there. This should generally be sufficient for someone else to duplicate the experiment and results. You will need to determine what conditions must be recorded that may affect your experiment. If I ask you to show me an effect you have observed, be able to reproduce the conditions. Detailed analysis need not be part of the lab book although some analysis is useful and you may wish to make a record here of all information that goes into your lab reports.

Do not erase anything. If you have errors, then cross out and correct or cross out and redo (justify this). Justify everything. If you retake data, and throw out old data, have a reason! Note anything unusual beyond the scope of the requested observations or tasks.
Note how your measurements are related to what you are trying to determine (i.e., why are you making these measurements). Do at least a preliminary analysis (i.e. if linear relation get a rough slope from first and last points....). Compare to standard or known results (if available). Discuss changes that might be made in experimental procedure.

Observations and Data collection can only be made while you are doing the experiment! YOU MUST ESTIMATE THE EXPERIMENTAL UNCERTAINTIES IN YOUR EQUIPMENT (HOW WELL CAN YOU READ, AND HOW RELIABLE ARE THE NUMBERS FROM YOUR EQUIPMENT).

When we do “hands on lab skills evaluation” Item 1 above—worth 25 pts, you must have your lab notebook with you. I will check and comment at that time .

LABORATORY REPORTS (word processed): I don't care what software you choose to use to make graphs, analyze, or to write with. We all have options such as Word, Excel, MatLab, and it is expected that you can use these (or will learn) sufficiently to analyze data, make graphs, and write reports. Ask if you need help, though most of you know these tools better than I do. Origin is highly recommended for graphing and line or curve fitting. If you do a line or curve fit (often required) you MUST include the statistical uncertainty resulting from the fit for each parameter (slope ±uncertainty in slope, or intercept….). LaTex is a popular documentation prep tool that you may wish to learn (I used once, but am not current).

The reports will follow an American Journal of Physics (AJP) format. I don't get terribly picky, but you must have neat appropriate information outlined below. (see the example links). You are expected to number formulas, include graphs and diagrams and handle the reporting of numbers properly (significant figures and use of scientific notation). Appropriate analysis is a must (different style is tolerated, but no tolerance for egregiously incorrect physics). **You may omit the double column format (difficult in word)---but all other format items should be followed very closely.**

You will need to make a judgment call regarding the physics literacy of the reader. You should anticipate advanced undergraduate physics majors. This means you can assume some common knowledge. You do not need to explain the use of "c" as the symbol for the speed of light. You may assume basic algebra and math skills. You should not clutter a paper with fluffy opinionated statements or with scratch paper like algebra steps. Be complete but brief.

You must have:

A title, names, date, and abstract along with other sections as discussed. Rather than the paper having the journal name in the footer, place "Advanced Physics PHYS4010 Spring 2019"

Abstract: Brief direct statement of results of the experiment with an equally brief indication of the general method used. You must include the important results with uncertainties. In the speed of light lab you must have c=###±##. For Schawlow's ruler you must have the wavelength with uncertainty. If you have a unique experimental method then you should give a very brief indication. If your results are long you may refer to an appendix or table in the paper. When possible you should indicate how far off your measurement is from the standard (eg. My result is 0.55 standard deviations [my uncertainties] from the accepted value). **Method and result.**

Introduction/Theory/Background: This can be one, two, three sections at your discretion. You may decide to add a section that includes "historical significance". Sufficient discussion and background material to relate the quantities being directly measured to those being determined in the experiment (how are your measurements used to arrive at results). Common results may be used directly. Special cases that apply to our experiment may require derivation of some expressions. **Proper citation of references if**

used (throughout). See sample papers for reference formats. This is where you would expect to see most numbered equations in the paper. In some cases this may be very brief. This (brevity) is not the case with the speed of light where there must be significant discussion regarding what you are tracking in time (critical).

Procedure: Complete description of how the measurements are taken. This should include a description of how the instruments are used. Do not reproduce manuals or initial alignment procedures unless critical to the specific experiment (as is the case with speed of light—WHY). Do not discuss common knowledge items such as turning a knob on an oscilloscope. Do discuss the features of the instrument you need to use for proper operation, and controls you use. If you use a HeNe LASER say so. If other properties matter, such as intensity or polarization properties, state those. You may include detailed information such as…."the longitudinal mode spacing leads to peaks produced by the laser are xxx+/- xxx ns apart". Some details may be best left to an appendix. You should include a diagram of the experimental set up (block figure sketch within Word is fine, and/or a picture with callouts and labels). You may need several setup figures. Do include relevant information (eg. the response time of the detectors is…….), don't include extraneous information (eg. it rained two days ago in southeast Asia). The reader must believe that you used a valid method and that there are no fatal flaws. Further if there is a new unique feature to your method, then you should indicate that. Goal: Be complete first, but be concise about it.

Results/Analysis/Discussion: Graphs, Tables, Charts. Report how the values you have obtained are found from the theoretical expressions and measurements. Report the method used to do a linear fit or canned program if used. Keep track of significant figures and uncertainties. Note! **Your analysis must include a determination of uncertainty in your final result**. This uncertainty represents the precision of the experiment. The uncertainty you cite in your final results should derive in a logical manner from the uncertainties in each of the quantities that you measured. This says how good you expect your experiment to be at measuring unknown quantities. How to handle uncertainties will evolve throughout the course. For most of the numbers you include in a lab report, there will be an uncertainty that you must write down.

Do not confuse the uncertainty discussed above (relating to how precise your experiment is) with accuracy. If you happen to know a standard value for the quantity you have measured then how close your result is to the standard is described by the accuracy. The accuracy relates your measured result to a known standard. There can be no mention of accuracy when measuring unknowns.

If you are dealing with known quantities you should discuss both the precision (uncertainty, standard deviation) as well as accuracy. For example if you measure a speed of light that is 2.91×10^8m/s $\pm 0.01 \times 10^8$m/s then the standard known accepted value is approximately nine standard deviations away from your result. This suggests that you may want to reconsider your uncertainty analysis, or look for unaccounted systematic errors in the experiment. Or your result may indicate that some new Nobel Prize winning physics is occurring and you need to investigate further why the speed of light differs in your experiment (uncertainty matters).

Discuss any possible systematic effects in the experiment, any improvements; compare to other methods of making similar measurements. Careful discussion and consideration of uncertainties will be given the largest consideration in determining grades on lab reports.

Conclusions or Discussion: If there is more you wish to add, do so here. This could be a place to discuss how the result will be used, or how it fits into science in general. Also one might discuss experimental improvements to be considered, or additional experiments that might be considered with similar set up. A comparison with known values might take place here. For example "my measurement of the speed of light in air is xxx standard deviations (my uncertainty) away from the standard value.

Remember that both writing and experimentation contain highly subjective elements and it is up to you to find ways to go beyond minimum performance.